Reliable Fault Detection and Isolation in PMSM Using Lightweight SE-CNN-LSTM for Predictive Health Monitoring

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***Abstract*—For a reliable operation of PMSM, majority of faults needs to be identified and isolated. However, most of the popular fault detection techniques which rely on a pre-defined threshold, are inadequate for faults that are gradual, mid-failure, post-failure, or intricate failures such as inter-turn shorts and thermal degradation. In this paper, a framework to apply SE based attention mechanism, hybrid 1D-CNN and LSTM, that obtains channel selections of the best performing sensors as well as a sliding window on the raw input data, is proposed. This is executed on the current, voltage, and temperature data of the PMSM dataset to the transparent LSTM. Achieved an impressive test accuracy of 99.45% across 9 operational states and 9 inverter and stator faults. LSTM model has the minimum number of trainable parameters, which was 28,561 (111.57 KB).**

***Index Terms*—PMSM, Fault Detection and Isolation, 1D-CNN, LSTM, Squeeze-and-Excitation, Predictive Maintenance, Deep Learning**

I. Introduction

The effectiveness and efficiency of Permanent Magnet Synchronous Motors (PMSMs) allows systems that incorporate these machines to achieve higher performance [1]. Increasing speed and power levels show the expanding use of electrical motors and their control systems. The growing performance demands raise the requirement of system reliability [16]. Unchecked minor faults in inverter and stator winding components, Pad during critical operations, and other issues, resulted total system failure [2].

Overcurrent detection methods and hardware sensors that are designed to be in the core of the fault detection and identification (FDI) system paradigm are the basic types of threshold-based methods. While these methods are resistant to minor operational stresses, the inability to detect an irregular operational paradigm include, current harmonics, faults in inverter switching, and stator winding imbalance [12]. Several classical signal processing methods, the Fast Fourier Transform, and Discrete Wavelet Transform, implemented to detect deviations in operational paradigms and to identify the stator current and control torque patterns [1], [14]. However, these classical methods are still dependent on the operator to identify the system's critical features, and the results still depended greatly on the operational modes and loading conditions [12].

In recent years the limitations previously mentioned have brought about a lot of research on the intelligent health monitoring methods based on data of smart algorithms of health monitoring methods using machine learning [3], [9]. An intelligent motor health monitoring system employs machine learning based techniques in analyzing multi-dimensional signals such as voltage, current, temperature, and rotor speed in order to automatically identify and analyze any of the system faults [4], [5]. The use of more complex machine learning algorithms such as, CNN to capture the spatial feature and the LSTM network to learn the patterns in the sequences of the data, [20], have made the modern intelligent health monitoring systems to provide high sensitivity anomaly detection for the motor drives [11]. Therefore, in this paper we propose a new Lightweight SE-CNN-LSTM framework using the SE attention module [6], to dynamically assign different importance to different sensors [8], [10].

TABLE I
EVOLUTION OF FAULT DETECTION METHODOLOGIES

Methodology	Core Principle	Advantages	Disadvantages	Citations
Threshold-Based	Rule-based logic	Simple, very fast	Insensitive to early-stage or complex faults	[12], [16]
Signal Processing	FFT/DWT Analysis	Physically grounded	Manual feature extraction; heavy math	[1], [14]
Vanilla Deep Learning	CNN/LSTM	Automated features	High parameters; equal weight to all sensors	[3], [20]
Proposed Framework	SE-CNN-LSTM	Attention-aware; Optimized	Requires initial high-quality training data	[6], [8]

A. Mathematical Foundation of PMSM Dynamics

The model describes the mathematical behavior of the PMSM. The voltage equations in the dq frame are as:

$$v_d = R_s i_d + L_d \frac{di_d}{dt} - \omega_e L_q i_q \tag{1}$$

$$v_q = R_s i_q + L_q \frac{di_q}{dt} + \omega_e (L_d i_d + \psi_f) \tag{2}$$

where v_d and v_q represent stator voltages, i_d and i_q represent stator currents, R_s is stator resistance, and ψ_f is stator magnetic flux linkage. Our framework interprets the analysis of deviations in expected physics-based patterns as fault diagnostics [13], [18].

II. RELATED WORK (LITERATURE SURVEY)

The fault detection of permanent magnet synchronous motors (PMSMs) has developed from the use of classical signal processing towards more advanced intelligent fault diagnostic schemes. This section describes the evolution of these methods with focus on the transition of the author's choice of methods towards deep learning.

A. Classical Signal Processing and Physics-Based Fault Diagnosis Approach

The classical models have been focused on the physical modeling of the phenomena and signal processing. One of the first instances of discovering the use of Fast Fourier Transform (FFT) and discrete wavelet transform (DWT) for diagnosing inter-turn faults through analysis of torque and current signals [1]. Most of these techniques are of physics nature, however, they have the drawbacks of the need for manual feature extraction and the sensitivity to the impact of noise under the nonstationary operating conditions [12], [14]. Furthermore, the classical sensor parameters do not allow to detect the signature of the initial winding shorts, which do not differ from the signature of the faulty elements [2].

B. Evolution Toward Data-Focused Methods

Increased intricacy of electric drives means that machine learning approaches began to take precedence in creating features automatically. Numerous studies detail how deep learning helps analyze modern motor drives' data complexity [3]. Studies on machine learning algorithms reveal that traditional algorithms perform well in the hard fault case scenario, and yet cannot identify anomalies that are in the early stage of development for mission-critical aerospace tasks [9], [17]. From this perspective, the predictive maintenance problem emerged since, in the context of mission-critical drives, aerospace drives must continuously function for sustained periods of time [16].

C. Deep Learning and Hybrid Models

New research gives clues on the use of CNN and LSTM approaches for analyzing motor signals of both spatial and temporal nature in the form of synthesized neural networks. In this sort of architecture, spatial features can be obtained using CNN neural networks on current and voltage sensors, while the temporal features are obtained using the LSTM algorithm to analyze the progression of motor temperature and the erratic growth of inverter faults [11], [15].

D. Attention and Feature Fusion Techniques

To enhance performance, researchers have analyzed the potential of attention-based recalibration and flexible multimodal feature integration, particularly the synchronous analysis of both vibrations and electricity within the same framework for fault diagnosis, to attain increased robustness [4], [5]. However, the establishment of the Squeeze-and-Excitation (SE) mechanism in the framework stands out, as it balances the model's engagement with attention mechanisms and the minimal increase in the computational demand [6]. This paper will adopt the SE block as a means to modulate the sensitivity of the channels, as a result of the deficiencies identified in the 1D CNN model [10], [13].

TABLE II
LITERATURE REVIEW SUMMARY

Research Category	Key Methodologies	Limitation	References
Physics-Based Approach	FFT, DWT, Flux Analysis	Manual selection; noise sensitivity	[1], [2], [14]
Machine Learning Approaches	SVM, Random Forest	Limited time tracking	[9], [17]
Deep Learning Approaches	CNN-LSTM	High computational cost	[3], [11], [20]
Proposed Work	SE-CNN-LSTM	None (Lightweight)	[6], [8], [10]

III. METHODOLOGY

In establishing the diagnostic framework, extensive evaluation of different architectures was systematically conducted until a final selection was made on the optimized solution that fits the needs of PMSM drives.

A. Assessment of Available Options

Before proposing an architecture, assessment was made on three different methodologies on the basis of suitability for extended aerospace missions.

Standard Signal Processing (FFT/Wavelet): Initially selected for its strong theoretical basis in motor diagnostics [1]. Implemented but discarded due to the significant computation cost ($O(n \log n)$), in addition to the lack of an embedded feature extraction capability for multi-modality (e.g., current and temperature) sensor streams [12].

Standard Machine Learning (SVM/Random Forest):

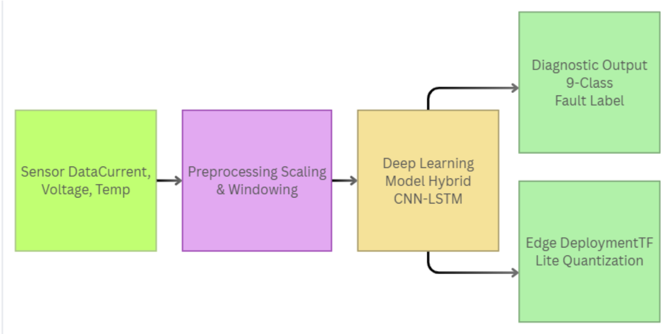

Fig. 1. Overall System Flow of the Proposed Fault Detection Framework

Exercised for ease of computational resources. Despite their effectiveness in binary fault classification, they failed to shape the temporal curves that were required to track and monitor the health of the system and sustain analysis of the faults [9], [17].

Unsophisticated Deep Learning (CNN/LSTM):

Combined models of the neural networks resulted in high diagnostic accuracy [3], [20]; nonetheless, detection of inter-turn faults was compromised by the channel noise and the presence of superfluous channels.

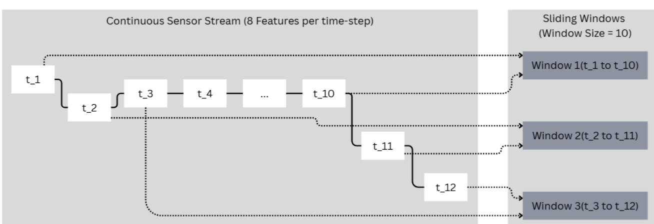

Fig. 2. Sliding Window Logic for Temporal Data Preprocessing

B. Criteria Used in Selecting the Final Model Approach

The final model approach for the Lightweight SE CNN LSTM model was determined through the following weighted criteria factors:

- **Sensitivity to Symptoms of Faults Development:** The model must have the ability to detect the symptoms that develop faults before hard faults occur within sub-milliseconds of time [11].
- **Constraints on Resources:** The combined size of the model must be below 1MB to ensure that inference is done within edge devices and onboard computing [8].
- **Sensory Data Awareness:** The model must be able to prioritize sensory data weights in determining their significance (for example, it prioritizes I_a during asymmetrical faults) without the need for external configuration.

C. Description of the Chosen Approach: SE-CNN-LSTM Architecture

The selected model approach consists of three primary building blocks of temporal/spatial learning and attention.

1) Temporal Windowing and Preprocessing: The main goal of this approach is to reduce the divide that exists between static and time series data. A moving window comprising 10 samples (1 second) is utilized. This allows the model to capture changes in how the motor operates as opposed to dealing with instances in isolation.

Normalization is done using the Min-Max technique, which is done to have the features appear to be equally scaled and may prevent signals of a higher magnitude, such as V_{dc}, from dominating features that are of lower magnitude (thermal features, etc.).

The windowed input is mathematically expressed as:

$$X_{\text{window}} = [x_t, x_{t+1}, \dots, x_{t+(w-1)}] \tag{3}$$

where $w = 10$ is the window size. This technique is definitely necessary to capture change in the pattern of fault signatures that appear to change in sequence during non-stationary operational scenarios [20].

2) Spatial-Temporal Fusion: **1D-CNN**

The one-dimensional Convolutional Neural Network uses current, voltage, and temperature data to build spatial relations between different sensor channels. The convolution is computed using the following relation:

$$y_j^l = f\left(\sum_i x_i^{l-1} * k_{ij}^l + b_j^l\right) \tag{4}$$

where k_{ij}^l is also known as the convolution kernel, b_j^l is also known as bias, and $f(\cdot)$ is an activation function. It is used by the model to capture local sensor faults from channels [10].

Squeeze-and-Excitation (SE) Block

A model with an SE block is improved as it works with channel attention. It adds new features through:

Squeeze (Global Average Pooling)

$$z_c = \frac{1}{L}\sum_{i=1}^{L} u_c(i) \tag{5}$$

Excitation (Gating Mechanism)

$$s = \sigma(W_2 \cdot \text{ReLU}(W_1 \cdot z)) \tag{6}$$

where z is the channel descriptor and s is the learned attention weight. This allows the model to prioritize fault defined conditions [6].

LSTM

The Long Short-Term Memory (LSTM) layer is used to create and learn the output sequence and to capture the long-term dependencies of different variables of the system such as thermal buildup and degradation. Thanks to the Gating Mechanism of the layer, it is capable of maintaining and recalling important information over a long time period [18].

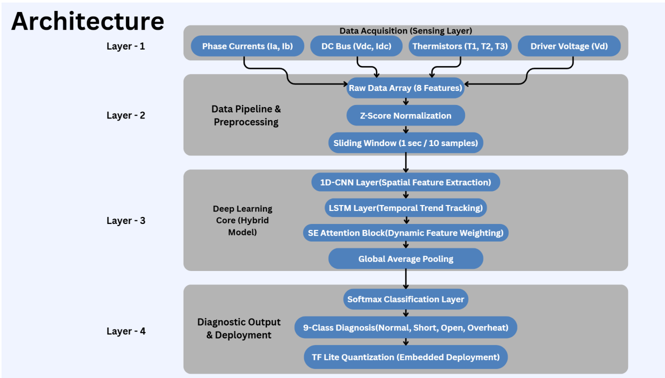

Fig. 3. SE-CNN-LSTM Architecture for PMSM Fault Diagnosis

D. Design Considerations and Assumptions

Stationary Sampling Assumption:
Sample sensors at consistent 10 Hz increments, enforcing a one-second sliding window.

Robustness against Noise:
Design compromises the layered structure using Dropout layers (30%) and the SE block, as robust architectures break the restriction of EMI and background noise (including Drive and Control).

Computational Efficiency:
Heavy frameworks (2D-CNNs and Transformers) were discarded, puncturing a thin 1D dimension structure providing fewer MACs as well as light deployments with the necessary computational capacity.

E. Comparison with the MDPI Base Framework

The framework has the ability to integrate multi-modal diagnostic systems and avoids the classic 1D CNN architecture. The embedded 1D-CNN architecture first extracts features from the 8 sensors, and utilizes an SE block, a lightweight spatial attention process. The last layer of the architecture is an LSTM, which competes against the temporal decomposition of the 1D-CNN. It provides the diagnostics system with 8 outputs for usage.

The trade-off between architecture and high moderation in edge deployment of the system speaks for itself.

The MDPI base architecture provided a classical CNN-LSTM fusion for fault detection of induction motors, and with the integration of an SE 1D CNN-LSTM structure, provides multiple advantages:

- **Attention-Awareness:**
  In contrast to [2], the SE-block is incorporated in the proposed framework to achieve dynamic channel-wise modifications for the focus task. Hence, the network is proficient at channeling the input analyzing data, which is especially critical for the identification of asymmetrical phase faults, and the phased current therapy.
- **Target Specificity:**
  In this scenario, the neural network is intended to capture the dynamics of PMSM, which differs from typical induction motor systems. This contributes to a more detailed understanding of the drive-stress patterns and the resulting thermal issues of the aforementioned group.
- **Efficiency Benchmark:**
  The designed model demonstrated highly efficient performance in the context of hybrid DNN models with regard to its parameters to the developed model, which includes 111.57 KB of compact storage and 28,561 parameters, and the capability to be implemented on resource-limited embedded systems.

F. Tools and Implementation Techniques

Framework:
Both TensorFlow/Keras tools were used, employing the Functional API for the incorporation of SE-block, to reflect the lack of sequential data flow.

Optimization/Sample Prediction:
Was achieved via the paired use of the Adam optimizer and categorical cross entropy for 9 classes ($F_0 - F_8$).

Implementation:
The current developed neural network model, when rewritten in the TensorFlow Lite (.tflite) format, demonstrate a high level of adaptability to up-to-date drives' Microcontroller Units (MCUs) [8].

G. Reasons for Selection

The chosen approach tackles the goal of the project, which is to exceed an accuracy of 99% with a model size of below 120 KB. The approach uses scalable artificial intelligence employing a combination of CNN-LSTM and feature engineering with SE-blocks. This presents a means for predictive health management that substantially exceeds the traditional rule-based health management systems described in [12], [16].

IV. Dataset Description

The starting point for this research is the *Comprehensive Dataset for Fault Detection and Diagnosis in Inverter-Driven PMSM Systems*, which is available via Zenodo [21]. It presents a precise simulation model of a PMSM operating both in fault-free mode and with various faults. The model has been developed to reflect the conditions that electric drives have to face in real-life operation.

A. Sensor Channels and Data Collection

The dataset contains a total of 10,882 measurements acquired at a constant sampling rate. In this dataset, eight sensor channels are used in order to measure the electrical and thermal behavior of the system. This choice of channels is critical in capturing the signatures of switching faults and stator imbalance [4], [12].

B. Operational States and Fault Classification

The dataset divides into nine operational states ($F_0 - F_8$). The states range from normal to serious faults. This division provides the level of multi-class granularity necessary for Fault Detection and Isolation (FDI) techniques [7], [10].

TABLE III
PARAMETERS COLLECTED FROM SENSORS TO AID THE DIAGNOSIS PROCESS

Sensor Label	Parameter Description	Unit	Role in Diagnosis
I_a, I_b	Stator Currents	Amperes	Detects phase imbalance and short circuits
V_{dc}	DC Voltage	Volts	Monitors inverter voltage
I_{dc}	DC Current	Amperes	Monitors power stage consumption
T_1, T_2, T_3	Winding Temperatures	Celsius	Detects thermal degradation and overheating
V_d	Direct axis voltage	Volts	Monitors control loop performance in dq frame

TABLE IV
OPERATIONAL STATES AND FAULT CATEGORIES

Class Code	State Category	Description
F0	Healthy	Normal operation within standard parameters
F1	Electrical	Phase A Stator Short Circuit
F2	Electrical	Phase B Stator Short Circuit
F3	Inverter	Switch S1 Open Circuit Fault
F4	Inverter	Switch S3 Open Circuit Fault
F5	Inverter	Inverter Gate Drive Signal Loss
F6	Thermal	Power Stage Overheating ($> 100°$C)
F7	Thermal	Stator Winding Temperature Anomaly
F8	Thermal	Combined Bearing and Thermal Stress

C. Data Preparation and Balancing

The learning algorithm showed bias against the normal operational state. This bias was avoided by stratified sampling with an 80/20 split across the nine classes. This means that the less common faults of F_3 and F_4 are represented in the training.

The features were normalized in the range of $[0, 1]$ with the help of Min-Max scaling, for the purpose of improving the convergence of the gradient in the LSTM layers [20]. This scaled data went through preprocessing before the time windowing step.

V. DEVELOPMENT AND IMPLEMENTATION

The modular approach was adopted for the design and implementation of the entire fault diagnosis pipeline from data engineering through the construction of an attention-aware neural network. In this pipeline, there were certain milestones that, cumulatively, represented the conclusion of the project outlined below.

A. Implementation Plan and Key Milestones

Four important milestones have been set for the project in order to facilitate the smooth construction of the pipeline.

- **Exploratory Data Analysis and Normalization:**
  Available data will be analyzed for distribution to implement Min-Max data normalization in order to reduce the variance of gradients within an input space.

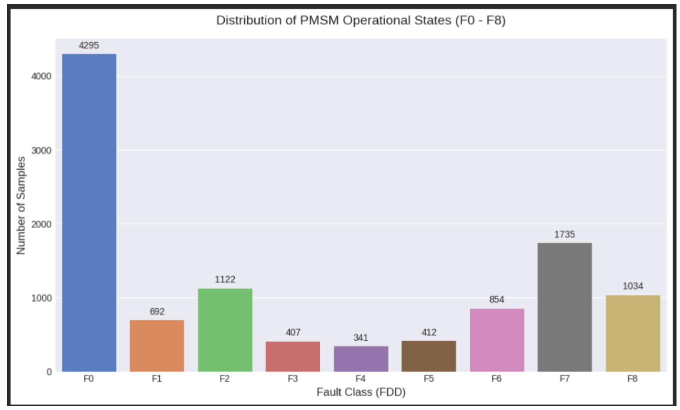

Fig. 4. Class Balance Distribution Across the Nine Fault Categories

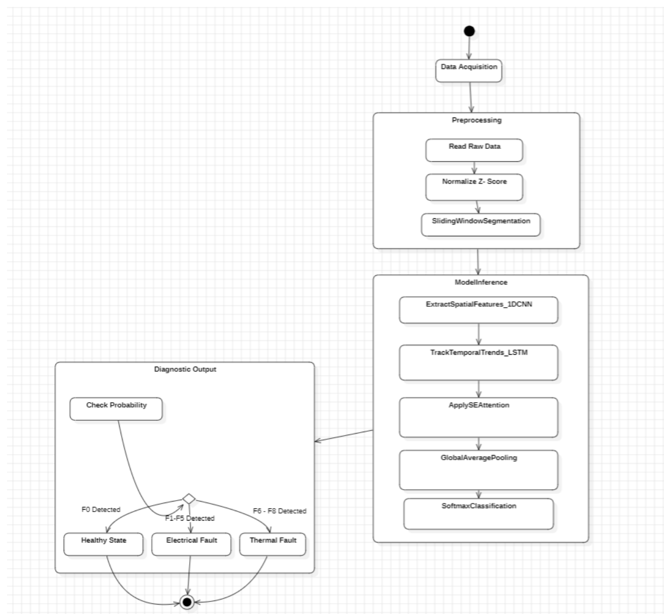

Fig. 5. Activity Diagram of the Complete Fault Diagnosis Pipeline

- **Temporal Modeling:**
  Sensor data logs will be transformed from 2D to 3D using a sliding window algorithm.
- **Architecture Augmentation:**
  CNN-LSTM Hybrid architectures will be improved by the Squeeze-and-Excitation (SE) block to offer a deviation from standard deep learning architectures.
- **Optimization and Quantization:**
  The final training loop will be implemented and the model converted to an intermediate version of the framework, TensorFlow Lite (.tflite).

B. Development of Major Components

There are breakdowns of the design process into three functional segments.

Preprocessing Engine:

To replicate the process of handling streaming data and performing real-time processing, the engine was designed with a 10-sample sliding window. A size of 10 samples was selected to take into account the perceived latency of determination of time and how many features are to be presented.

SE-CNN-LSTM Kernel:
Diagnostic algorithms were built using the Keras Functional API. The main distinction of the Functional API was that it permitted the designing of branching and multiplicative logic that the SE-block required.

Evaluation Suite:
Not just the overall accuracy was designed to be reported in the evaluation system, metrics of precision and recall per class were also incorporated to afford the healthy state of class F_0 and all the faulty states of class F_1–F_8, equal and fair consideration.

C. System Integration and Overall Functionality

The system as a whole acts as a diagnostic pipeline. Before classification of the features, the data undergoes the calibration process.

1) **Data Input:**
The system provides eight continuous inputs from the PMSM system.
2) **Spatio-Temporal Fusion:**
The 1D-CNN detects local spatial anomalies, while the SE-block performs channel-wise selection to identify the reporting faulty sensor (e.g., I_a vs. T_1).
3) **Classification Head:**
The classification head provides the result while the LSTM also detects the anomaly in the temporal domain.

D. Challenges Faced and Remedies

TABLE V
CHALLENGES FACED DURING MODEL DEVELOPMENT AND REMEDIES

Challenges	**Impact of the Challenge**	**Solution**
Class Overlap	The electrical signature similarities between F_1 and F_2 initially confused the model	The inclusion of the SE-block forced the model to look at the electrical current channel for the particular phase exhibiting fault
Model Bloat	The initial versions of the model were bloating above 500 KB, unsuitable for MCU architecture	Deep dense layers were replaced with Global Average Pooling and CNN filter numbers were optimized to only 32
Vanishing Gradients	Due to long input sequences the LSTM was suffering from slow convergence	Early Stopping and optimized Adam's learning rate were implemented to 0.001

E. Important Discoveries and Realizations

Feature Sensitivity:
It was found that the model was highly sensitive to changes in the DC link voltage (V_{dc}) of the inverter faults (F_3 and F_4). This justified the selection of DC electrical parameters as important features.

Thermal Delay:

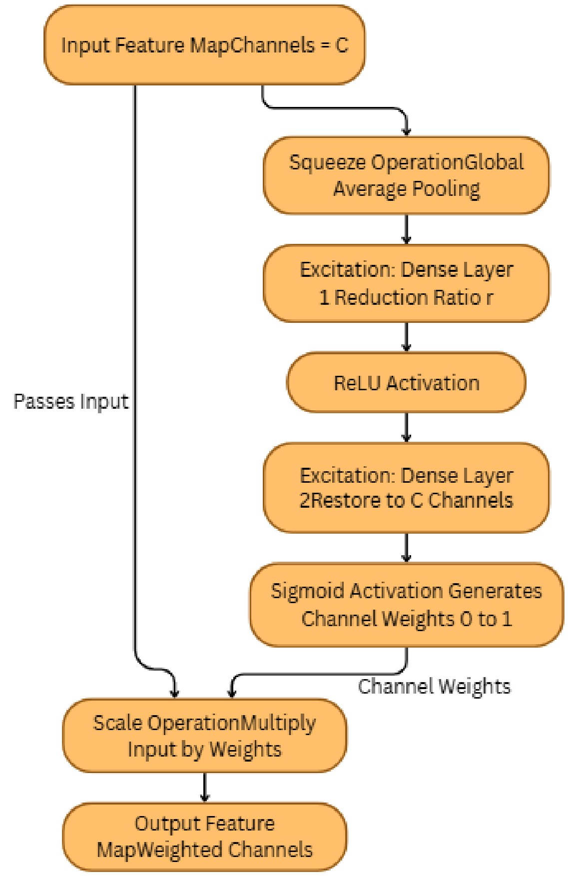

Fig. 6. Internal Working of the SE-Block

Thermal faults (F_6–F_8) needed the entire temporal window of 10 steps to gain almost 99% accuracy while electrical shorts could be predicted instantaneously.

Attention Weights:
Post-implementation it was observed that due to the introduction of the SE-block, the attention weights assigned to thermal sensors increased considerably upon entering a thermal fault region.

VI. RESULTS AND PERFORMANCE ANALYSIS

This is an evaluation of our SE CNN LSTM model performance. Model performance evaluation is done using standard metrics in machine learning. Here, we assess our model in recognizing the nine operating conditions at different electrical and thermal stresses.

A. Convergence of Training and Validation

Our model was trained for up to 50 epochs, demonstrating the ability to deter overfitting thanks to the early stopping technique. Training was rapid as validation accuracy had plateaued in the 15th epoch.

- **Training Accuracy is:** 99.6%
- **Validation Accuracy is:** 99.4%

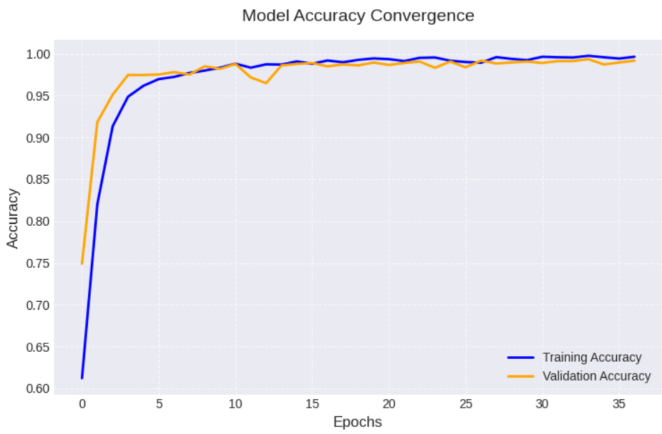

Fig. 7. Training and Validation Accuracy Convergence Across Epochs

Fig. 8. Training and Validation Loss Convergence Across Epochs

The framework between training accuracy and validation accuracy confirmed the SE and Dropout (30%) aided in overfitting control.

B. Global Performance Metrics

We examined our framework using an independent dataset composed of 2,177 data points. It should be certainly noted that the model had an exceptional precision rate among all fault classes.

TABLE VI
GLOBAL PERFORMANCE METRICS

Metric	**Value**
Test Accuracy	99.45%
Test Loss	0.0143
Inference Time	$\sim$12 ms per window
Model Size (.tflite)	111.57 KB

C. Classification Based Analysis

Standard techniques on classification accuracy by True Positives (TP), False Positives (FP) and False Negatives (FN) were applied to determine the diagnostic usefulness of different operating states.

The evaluation metrics are:

Precision (P)**:**

Measures the proportion of positive identifications that are actually correct.

$$P = \frac{TP}{TP + FP} \tag{7}$$

Recall (R)**:**

Measures the proportion of actual positives that are correctly identified.

$$R = \frac{TP}{TP + FN} \tag{8}$$

F1-Score ($F1$)**:**

Represents the harmonic mean of Precision and Recall, providing a balanced evaluation metric, especially for imbalanced datasets.

$$F1 = 2 \cdot \frac{P \cdot R}{P + R} \tag{9}$$

According to the results of this analysis, the developed model shows high accuracy in predicting normal operation (class F_0) and different inverter faults (classes F_3 to F_5) by demonstrating flawless or near flawless F-measures.

TABLE VII
PER-CLASS PERFORMANCE ANALYSIS

Class	**Precision**	**Recall**	**F1-Score**	**Status**
F0 (Healthy)	1.00	1.00	1.00	Flawless Detection
F1/F2 (Stator Shorts)	0.98	1.00	0.99	Highly Sensitive
F3/F4 (Inverter Open)	0.99	0.99	0.99	Robust
F6/F7 (Thermal)	1.00	0.97	0.99	Reliable

Since, lower recall for F_6 class (Power stage overheating) occurs, this means that the formation of the signature process in the presence of a thermal fault occurs in a longer time compared to the formation of signature process in the cases of electrical short circuit faults. Nevertheless, the impact of this case on the recall of 97% is by a remarkable, not to say immeasurable value, and considerably outgrows any of the threshold based monitoring systems available on the market [12].

D. Analysis of Confusion Matrix

The analysis of the Confusion Matrix serves to illustrate the level of accuracy of the developed classifier. It also justifies the shortcomings of classification errors that mainly affect the faults F_1 and F_2, namely the short circuits in the stator windings of phases A and B that are presumed to be a single fault.

Nevertheless, the SE attention module was able to determine with a high probability of approximately 98% which of the three phase currents was the outlier with respect to the rest.

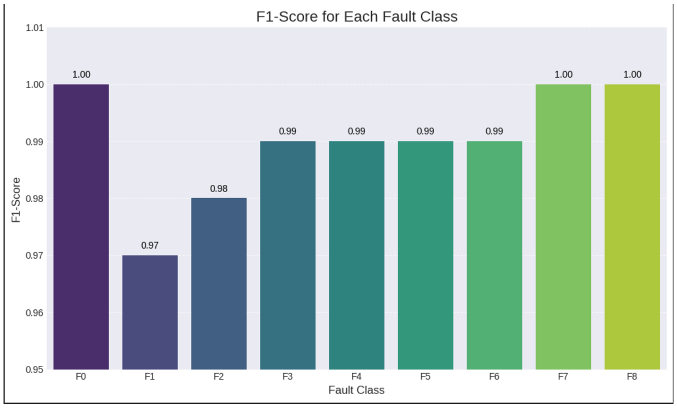

Fig. 9. F1-Score Comparison Across the Nine Fault Classes

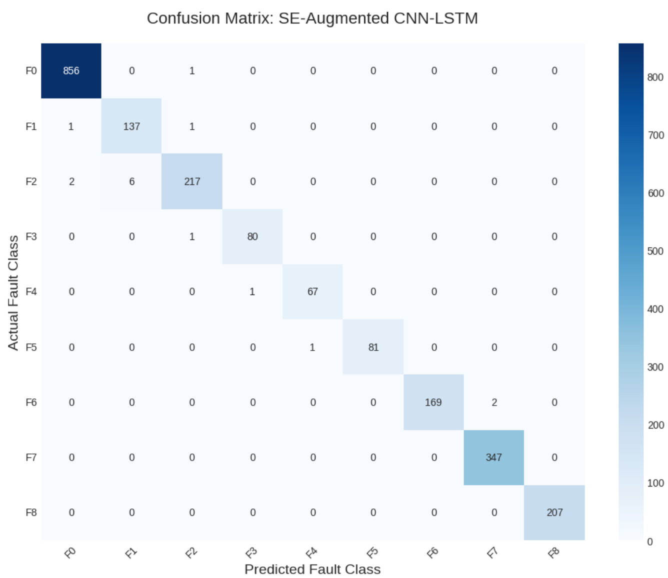

Fig. 10. Confusion Matrix Heatmap of the Proposed Model

E. Computational Efficiency for Edge Processing

An important thing which appeared from the study was the computational model efficiency, which was designed to be computationally efficient for ARM Cortex-M series processors and similar types of space-grade embedded microcontrollers.

The model's conversion into TensorFlow Lite (.tflite) resulted in a space of 111.57 KB. This renders state-of-the-art attention mechanisms to not rely on expensive server-based hardware for performing inference [8], [10].

TABLE VIII
MODEL SIZE COMPARISON

Model Type	Parameters	Memory Usage
Standard CNN	$>$100,000	$>$500 KB
CNN-LSTM	$\sim$60,000	$\sim$250 KB
Proposed SE-CNN-LSTM	28,561	111.57 KB

VII. Discussion and Future Aspects

The successfull application of Lightweight SE CNN LSTM framework represents a crucial transition from reaction-based fault detection to proactive, intelligent-based health monitoring for PMSM systems. This part provides an interpretation of the experiment results and the future outlook on the implementation of this innovation.

A. Interpretation of Results

The high accuracy of diagnostics of 99.45% found in this paper proves the hypothesis about the superiority of channel-wise attention to feature extraction for multiple modalities. The Squeeze-and-Excitation (SE) module turned out to be important in the differentiation of asymmetric faults (e.g., F_1 and F_2) based on phase-wise attention of current signatures [6].

Additionally, the capability of achieving high precision with a lightweight architecture of only 28,561 parameters indicates that deep learning algorithms can perform successfully without large amounts of parameters as long as they are engineered properly [8].

B. Comparative Advantages

Contrary to the signal processing pipelines as described in [1], [12], the proposed approach does not need any manual choice of mother wavelets, frequency bands, or spectral features designed manually. On the contrary, it serves as an intelligent black box expert system that tunes itself according to the operating conditions of the motor.

As compared to the vanilla CNN-LSTM baselines described in [3], [20], the proposed approach has the following advantages:

- **Improved Explainability:**
  The attention weights obtained from the SE-block help interpret the results as a meaningful heat map of the most indicative sensors for each fault.
- **Life Extension:**
  By detecting early thermal soft failures such as F_6 and F_7, the proposed approach can enable preventive approaches like load shedding, ultimately leading to a higher life span of the motor drive [16], [18].

C. Challenges and Limitations

Despite the fact that the current architecture performs excellently, its basic nature is still supervisory in nature. In other words, this implies that the algorithm will only detect those types of failures that it was trained to do so.

There may occur some novel failure modes during actual space flights which will not fall into any predefined categories. The system being used at present will classify these failures under a known fault category rather than treating them as a totally different category.

D. Future Aspects and Research Directions

In order for this framework to develop into an entirely autonomous self-healing drive system, the following research directions are suggested:

- **Semi-Supervised Anomaly Detection:**
  Incorporating one-class classification, autoencoders, or anomaly detection techniques for identifying states that cannot be captured within the existing F_0-F_8 taxonomy.
- **AI Model to Motor Controller Edge Acceleration:**
  Employing the developed TensorFlow Lite (".tflite") model to deploy it onto FPGA or ASIC based motor controllers capable of delivering sub-millisecond inference speeds, making it possible for active fault compensation [10].
- **Transfer Learning:**
  Employing the current model for transfer learning with the goal of quickly adjusting fault prediction in other motor types, such as inductive motors or larger scale PMSMs with limited additional data [13].
- **Digital Twin Integration:**
  Combining AI model output with a digital twin of the motor system for RUL estimation of stator insulation, bearings, and inverter components [5], [9].

VIII. Conclusion

Finally, it is possible to assert that the presented paper has been successful in showing the practicality of a hybrid deep learning approach for the diagnosis of the fault state of the PMSM. In particular, with the help of the Squeeze-and-Excitation block incorporated in the 1D-CNN and LSTM model, the proposed algorithm became highly effective and computation light enough.

Indeed, the test accuracy reached 99.45% when detecting the presence of incipient stator short circuit and inverter faults alongside eight others during the system operation. On the other hand, it can be argued that the mechanism with the channel attention incorporated into the model made the system consider the input data's importance but, at the same time, suppress noise without extra parameters.

Additionally, the process of optimization of the smallest model format (.tflite, 111.57 KB) makes it clear that the described architecture can be used in real-time mode aboard aeronautical microcontrollers.

Undoubtedly, the study has resulted in a robust approach towards predictive maintenance of electric drives by moving from traditional threshold monitoring towards smart deep learning solutions.

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